

Month 1 - Python Fundamentals

Topics:

- Introduction to Python, Variables, Data Types (int, float, string, boolean)
- Operators (arithmetic, comparison, logical)
- Control Flow (if, elif, else statements)
- Basic Input/Output

Project:

- A simple calculator that performs basic arithmetic operations.

Resources:

- [Python Tutorial for Beginners - freeCodeCamp](<https://www.youtube.com/watch?v=rfscVS0vtbw>)
- [Official Python Tutorial](<https://docs.python.org/3/tutorial/>)

Practice:

- [HackerRank - Python Basic Data Types](<https://www.hackerrank.com/domains/python/py-basic-data-types>)
- [Edabit - Python Beginner Challenges](<https://edabit.com/catalog/python-beginner-exercises>)

Month 2 - Data Structures & Loops

Topics:

- Lists, Tuples, Dictionaries, Sets
- Loops (for, while)
- List Comprehensions

Project:

- A to-do list application that allows adding, removing, and viewing tasks.

Resources:

- [Python Data Structures - GeeksforGeeks](https://www.geeksforgeeks.org/python-data-structures/)
- [Learn Python - Data Structures - Real Python](https://realpython.com/python-data-structures/)

Practice:

- [HackerRank - Python Strings](https://www.hackerrank.com/domains/python/py-strings)
- [Coderbyte - Basic Python Challenges](https://coderbyte.com/challenges/python)

Month 3 - Functions & Modules

Topics:

- Defining and calling functions
- Function arguments (positional, keyword, default)
- Scope (local, global)
- Importing and using modules (math, random, os)

Project:

- A password generator that creates strong, random passwords.

Resources:

- [Python Functions - Real Python](https://realpython.com/learning-python/)
- [Python Modules - Python Official Docs](https://docs.python.org/3/tutorial/modules.html)

Practice:

Katas](<https://www.codewars.com/kata/search/python?q=8-kyu&beta=false>)

- [HackerRank - Python Functionals](<https://www.hackerrank.com/domains/python/py-functionals>)

Month 4 - File Handling & Error Handling

Topics:

- Reading from and writing to files
- `try-except` blocks for error handling
- Common exceptions

Project:

- A simple script to read data from a CSV file, perform some calculations, and write results to another file.

Resources:

- [Python File I/O - W3Schools](https://www.w3schools.com/python/python_file_handling.asp)
- [Python Error Handling - Real Python](<https://realpython.com/python-exceptions/>)

Practice:

- [HackerRank - Python IO](<https://www.hackerrank.com/domains/python/py-io>)
- [LeetCode - Easy Python Problems (focus on data manipulation)](<https://leetcode.com/problemset/all/?difficulty=easy&page=1&listId=o3p2f0>)

Month 5 - Object-Oriented Programming (OOP) Basics

Topics:

- Classes and Objects
- Attributes and Methods
- Inheritance
- Polymorphism

Project:

- A simple inventory management system using classes (e.g., Item class, Inventory class).

Resources:

- [Python OOP Tutorial - freeCodeCamp](https://www.youtube.com/watch?v=Ej_02ICOXyl)
- [Object-Oriented Programming (OOP) in Python 3 - Real Python](https://realpython.com/python3-object-oriented-programming/)

Practice:

- [HackerRank - Python Classes](https://www.hackerrank.com/domains/python/py-classes)
- [Exercism - Python Track (OOP focused exercises)](https://exercism.org/tracks/python)

Month 6 - Introduction to Libraries & Job Readiness

Topics:

- Popular libraries: `requests` (for APIs), `pandas` (for data analysis basics)
- Introduction to Virtual Environments (`venv`)
- Basic Git and GitHub for version control
- Portfolio building tips and common interview questions.

Project:

- Fetch data from a public API (e.g., a weather API or a movie database API) using `requests`,

process it with `pandas` (if applicable), and present it in a readable format.

Resources:

- [Requests Library Documentation](https://docs.python-requests.org/en/latest/)
- [Pandas Documentation](https://pandas.pydata.org/docs/)
- [Git Handbook](https://guides.github.com/introduction/git-handbook/)
- [Python Interview Questions for Beginners](https://www.simplilearn.com/python-interview-questions-and-answers-article)

Practice:

- [HackerRank - Problem Solving](https://www.hackerrank.com/domains/python/python-problem-solving)
- [LeetCode - Explore Cards (Start with "Learn SQL", "Top Interview Questions")](https://leetcode.com/explore/)